Torque-speed and efficiency report

Validation IPM 14p Flat Buried — Large. Built on demand from the design's flux-linkage map and the drive limits set on the Torque-speed tab, with the same calculation the tab uses. The run's own report records the single solved operating point.

Key results

Peak torque	105.721 N m
Base speed	726.795 rpm
Peak power	9.283 kW
Top speed reached	2102.388 rpm
Best efficiency	97.09%

Status

Qualified outputs pass a benchmark gate against a reference; experimental outputs have none yet.

Torque-speed envelope	Experimental: Not yet compared with measured torque-speed data. Interpolated from the flux-linkage map with sinusoidal current; inverter drops and dead time are ignored.
Efficiency map	Experimental: Not yet compared with measured efficiency data. MTPA control: the smallest current for each torque within the voltage limit, interpolated from the flux-linkage map; torque is electromagnetic. Copper loss at the winding temperature; iron loss scaled from the map's reference speed, hysteresis with frequency and eddy current with its square.
Map flux linkage, MTPA trajectory and inductances	Experimental: The d/q gates cover 12 operating points, not a map's full range of currents and angles, and do not bound the inductances.
Phase resistance and copper loss	Experimental: No wire is set, so the wire is sized to fill the slot to its limit: an estimate, not the resistance of a real winding. Set the wire for a resistance the phase-resistance gate covers.

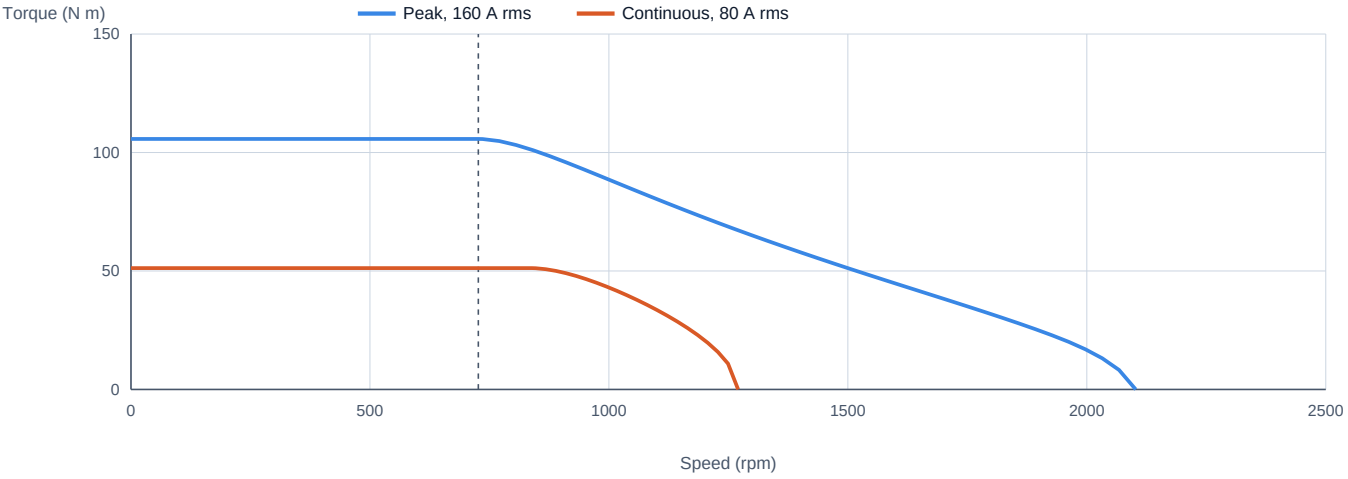
Drive inputs

Peak current	160 A rms (226.274 A peak)
Continuous current	80 A rms (113.137 A peak)
DC bus	48 V, as entered
Phase voltage limit	27.713 V peak (modulation index 1.155)
Top speed	5400 rpm
Drive control	MTPA, then field weakening (least current per torque)
Phase resistance	0.002 ohm at 20 degC, experimental
Mechanical loss	0 W
Flux-linkage map	7 currents up to 226.274 A peak x 10 angles (0 to 90 deg), 24 rotor positions each, normal mesh, standard grid; iron loss at 1800 rpm
Map scope key	f6596cd700479cfb18320ef25cf8bf6c52f734e5121049c4d32964feb7be5603
coilEM	3ce46ed1ba1bcf19b422e504248917843dcf22cf

Torque and power against speed

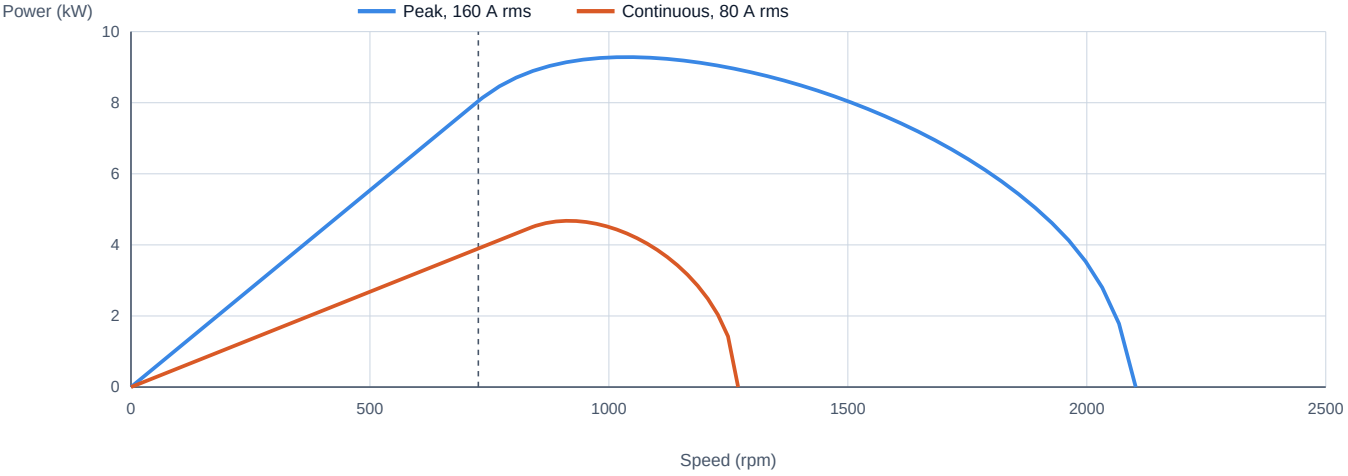
Torque against speed

Experimental



Power against speed

Experimental



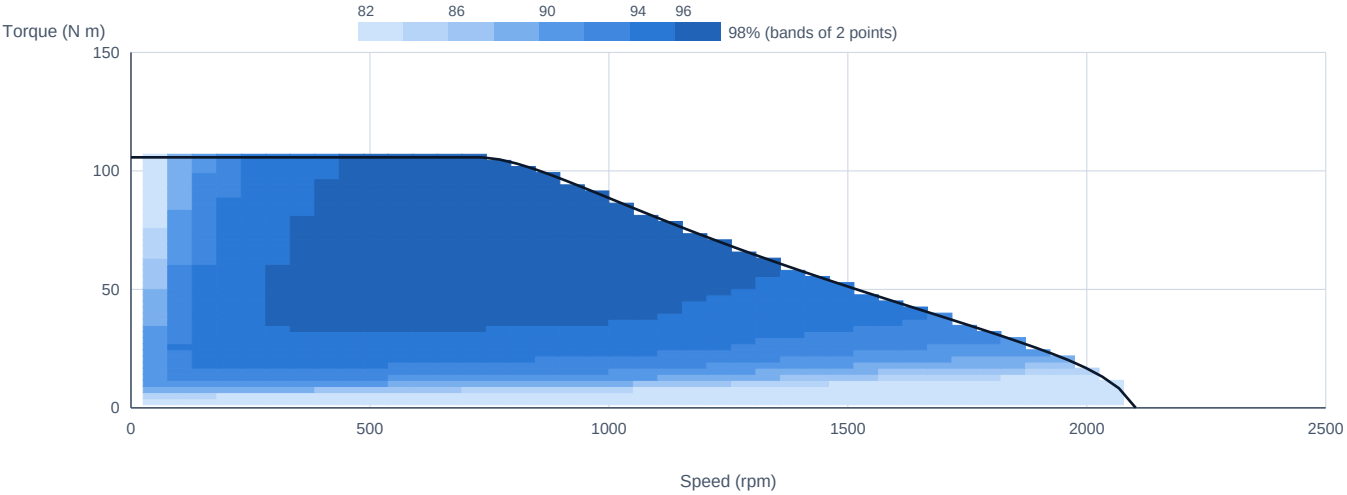
Peak curve at selected speeds (every sample is in the performance CSV)

Speed (rpm)	Torque (N m)	Power (kW)	Current (A rms)	Angle (deg)	Voltage (V peak)	Region
0	105.721	0	160	12.8	0.397	MTPA
210.239	105.721	2.328	160	12.8	8.263	MTPA
385.438	105.721	4.267	160	12.8	14.859	MTPA
595.677	105.721	6.595	160	12.8	22.775	MTPA
735.836	105.677	8.143	160	14.315	27.713	Field weakening
946.075	92.986	9.212	160	38.138	27.713	Field weakening
1121.274	78.644	9.234	160	49.764	27.713	Field weakening
1331.512	62.736	8.748	160	59.637	27.713	Field weakening
1506.711	50.755	8.008	160	66.106	27.713	Field weakening
1716.95	37.201	6.689	160	72.872	27.713	Field weakening
1892.149	25.451	5.043	160	78.411	27.713	Field weakening
2102.388	0	0	160	89.981	27.713	Field weakening

Efficiency map

Efficiency map, MTPA, then field weakening

Experimental

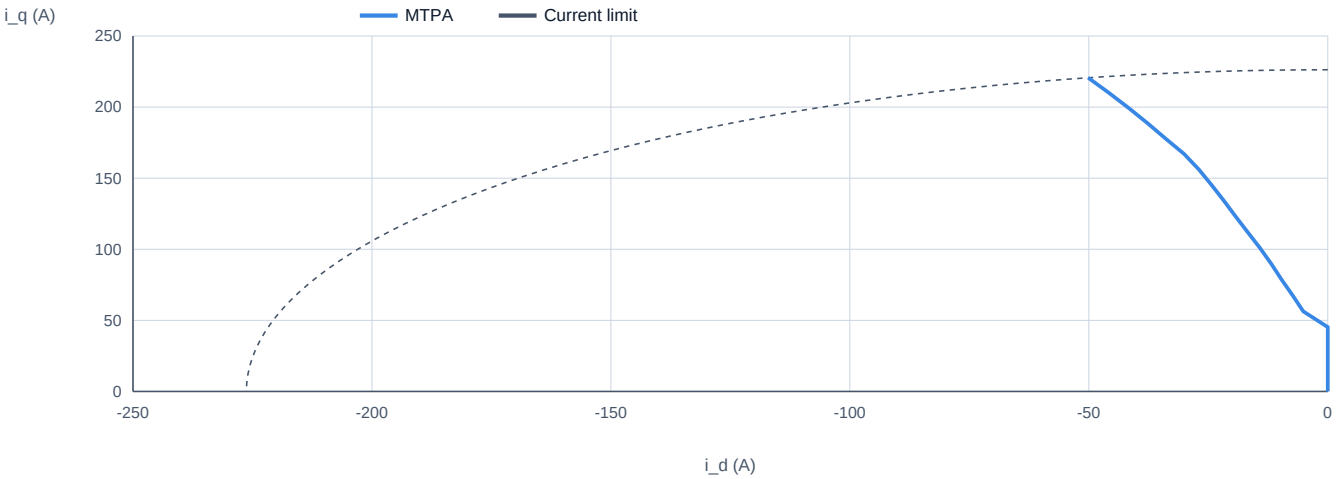


Efficiency = shaft power / input power, with shaft power = T x omega - iron loss - mechanical loss and input power = T x omega + copper loss. Cells above the peak curve are outside the drive's reach. The current and angle of every cell are in the performance CSV.

MTPA trajectory and flux linkage

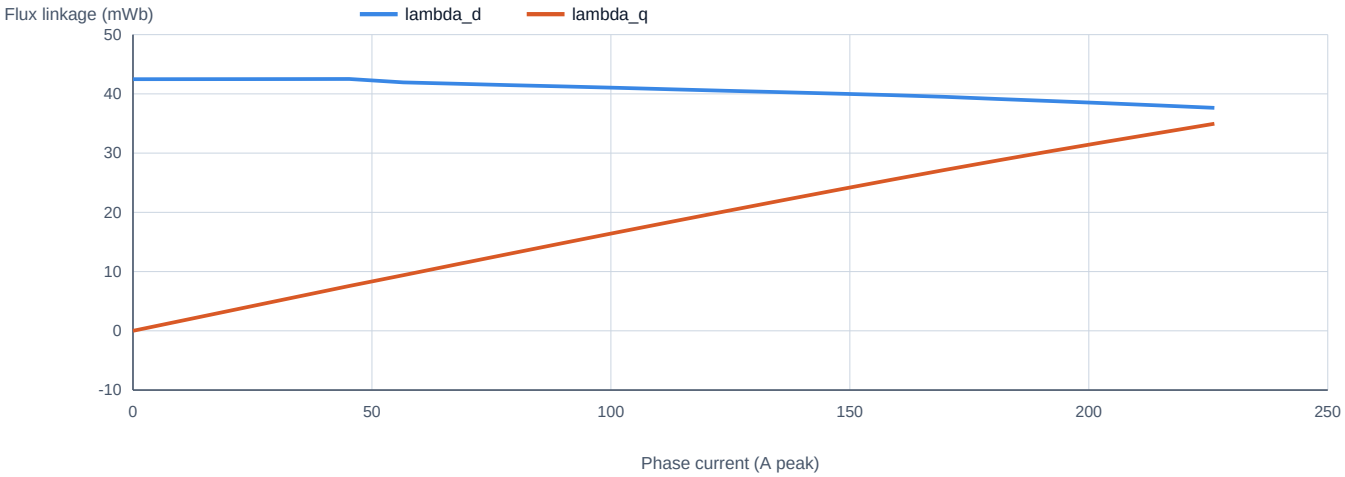
MTPA trajectory in the id-iq plane (peak A)

Experimental



Flux linkage along the MTPA trajectory

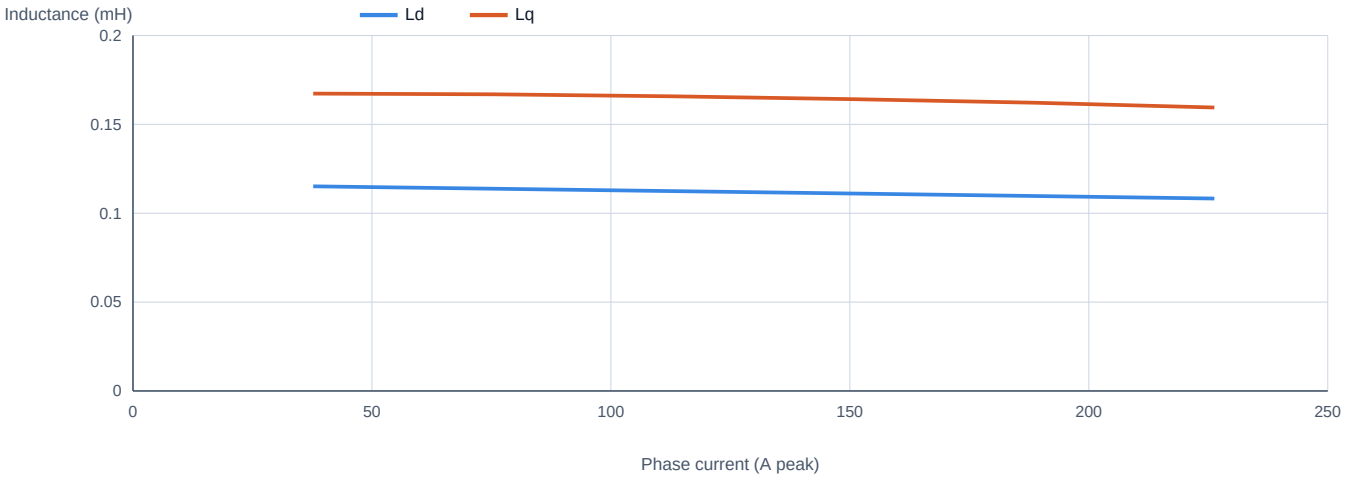
Experimental



Apparent inductance

Apparent inductance against current

Experimental



Current (A peak)	Current (A rms)	Ld (mH)	Lq (mH)
37.712	26.667	0.115	0.167
75.425	53.333	0.114	0.167

Current (A peak)	Current (A rms)	Ld (mH)	Lq (mH)
113.137	80	0.112	0.166
150.849	106.667	0.111	0.164
188.562	133.333	0.11	0.162
226.274	160	0.108	0.159

Notes

- FEM torque is 100.1-100.9% of the d/q torque across 53 map points.

- Ld from the 90 deg points, (lambda_d - lambda_pm) / i_d; Lq from the 0 deg points, lambda_q / i_q. Both are secant values and include saturation.

- The MTPA angle is read off the interpolated map, whose angles are 10 deg apart at most. At low current, where torque hardly changes with angle, it is approximate; a finer map grid gives a smoother trajectory.

- Not yet compared with measured torque-speed data. Interpolated from the flux-linkage map with sinusoidal current; inverter drops and dead time are ignored.

- Not yet compared with measured efficiency data. MTPA control: the smallest current for each torque within the voltage limit, interpolated from the flux-linkage map; torque is electromagnetic. Copper loss at the winding temperature; iron loss scaled from the map's reference speed, hysteresis with frequency and eddy current with its square.

Known limitations

- Interpolated from a flux-linkage map (Catmull-Rom over current and current angle); nothing is extrapolated beyond its currents and angles. Not yet compared with measured data.

- Steady state with sinusoidal current. Inverter voltage drops, dead time and the voltage behind the iron-loss resistance are left out.

- Torque is electromagnetic. Shaft output subtracts iron loss and the mechanical loss entered for the design.

- Iron loss is scaled from the map's reference speed: hysteresis with frequency, eddy current with its square.